

COMS5022A

Programming for Data Scientists

Solution Report



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Declaration of Group Members:

We as Group 16 declare that this report is of our own, unaided work. It is being submitted for the Postgraduate Diploma in Data Science for the module **(Programming for Data Science Introduction)** factuality of Science at the University of the Witwatersrand, Johannesburg.

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Nomenclature

Note: Only acronyms used in this report are listed below.

HR – Human Resources

NaN – Not a Number

Pytest – Python Testing Framework

1. Introduction

This project focuses on setting up a basic HR analytics system using Python. The system processes employee data, identifies data quality issues, performs cleaning operations, and computes key HR metrics.

The dataset is structured as a matrix, where each inner list represents an employee record. Each column corresponds to a specific attribute such as department, age, salary, and performance rating.

The main objective is to transform raw and uncleaned data into a clean dataset that supports accurate analysis. Core programming concepts including loops, conditionals, lists, sets, and dictionaries are demonstrated, with no external libraries being used, as per project requirements

2. Data Cleaning (Part A)

Data cleaning ensures the dataset is reliable before performing analysis. Poor data quality leads to incorrect outputs and runtime errors. (Witten et al., 2011; Provost and Fawcett, 2013)

2.1. Remove Null Salaries

Problem Statement:

- Some of the employee records contained invalid salary entries, where the word 'nan' was recorded in the place of actual salary figures.

Why It Matters:

- In the metric calculations, the salary entries are used to calculate average salaries for given age ranges. The invalid or uncleaned 'nan' values will break arithmetic operations and produce invalid results.

Approach:

- Iterate through each record in the dataset
- Access the salary field using index 4
- Use `math.isnan()` to detect NaN values
- Skip records with invalid salary values

- Store valid records in a new list
- Return the cleaned dataset

Outcome:

- The dataset contains only valid numeric salary values, ensuring accurate calculations in later stages.

2.2. Standardise Departments

Problem Statement:

- The various Departments have inconsistent names where some consist of a combination of lower and upper case alphabets, some are only lower case alphabets while some are only upper case alphabets.

Why It Matters:

- With Python being a case-sensitive programming language, two alphabetically-identical names would be considered as two different entries by Python if they are of different cases.

Approach:

- Loop through each record
- Access department field at index 1
- Convert each value to lowercase using string methods
- Update values directly in the dataset

Outcome:

- All department names are uniform, which ensures correct grouping and aggregation.

2.3. Remove Invalid Performance Ratings

Problem Statement:

- Some records contain performance ratings outside the valid range of 0 to 5.

Why it matters:

- Invalid ratings, which are values outside of the given range will give a false representation of the performance of the departments.

Approach:

- Iterate through dataset
- Access performance rating at index 6
- Check if value lies within 0 and 5 inclusive
- Remove records that fall outside this range
- Return filtered dataset

Outcome:

- Once the invalid performance ratings have been removed, the dataset will give a true representation of the performance of each department.

2.4. Fix Hire Dates

Problem Statement:

- Hire dates appear in multiple formats and include invalid values.

Why it matters:

- The company was established in January 2015 therefore any hire dates before that would be inaccurate. Any records after December 2025 would be considered to be in the future. Invalid date values also include days that are less than 1 and greater than 31. Months that are less than 1 or greater than 12 are also considered invalid. For years that were not leap years the 29 February date would also be considered invalid. Invalid date formats that are not consistent with the system dates will not be translated.

- **Approach:**

- Formatting

- Loop through each employee record.
- Identify dates in DD/MM/YYYY format
- Rearrange components into YYYY-MM-DD format
- Update dataset in place

- Validation

- Extract year, month, and day from each date
- Check year is between 2015 and 2025
- Check month is between 1 and 12
- Check valid day ranges
- Apply leap year logic for February
- Remove records with invalid dates

Outcome:

- All dates follow a consistent format and fall within valid ranges.

3. Metrics Calculation (Part B)

After the dataset was cleaned, several HR metrics were calculated. These metrics help provide insights into workforce composition, retention, performance, and compensation trends.

- Identification of unique departments in the organisation.
- Calculation of gender distribution per department as percentages.
- Calculation of average employee age per department.
- Determination of overall employee retention rate.
- Calculation of turnover rate per department.
- Calculation of average salary within a specified age range, including edge case handling.
- Calculation of average performance rating by department within a training hours range.

3.1 Unique Departments

The unique departments metric carried out the function of identifying all the distinct departments within the organisation. This metric serves as the basis for grouping other analyses and ensures that calculations such as averages and distributions are correctly segmented.

The following approach was used to identify the unique departments:

The implementation extracts department values from index 1 of each record, removes duplicates by inserting the values into a set, and returns the resulting set of all the unique department names.

3.2 Gender Distribution

Gender distribution calculations were taken as a percentage of gender per department allowing for a fair comparison between departments of different sizes. This metric provides information into workforce diversity and also reveals potential imbalances within departments. The implementation counts the distinct gender values within every department, it then stores the counts in a dictionary, the values are then converted to percentages and returned as a nested dictionary structure.

3.3 Average Age per Department

The average age per department metric provides a summary of the age distribution of employees within each department. This metric provides an in-depth understanding of various experience levels and workforce planning. It is also useful in the identification of departments that have ageing employee profiles. The implementation groups employee ages by department, computes the average age for each group, and returns the results in a dictionary.

3.4 Retention Rate

The retention rate metric was calculated as a percentage of active employees in relation to the total workforce. It helps provide a high-level visual of employee stability within the organisation. The implementation counts employees with an “Active” work status, continues to count the total number of employees and uses the values to calculate the percentage. The final percentage values are then rounded off to two decimal places.

3.5 Turnover Rate per Department

The departmental turnover rate is a measure of the frequency at which employees resign within a department and it helps with the identification of areas that may require policy or managerial intervention. The implementation counts the number of employees who have resigned in each department, it then divides the value by the total number of employees in that department and then multiply the ratio by 100 to get a percentage. The results are then stored in a dictionary.

3.6 Average Salary Age per Range

The average salary for a given age range metric serves as a support for the analysis of the various compensation levels for specific age groups. It also accounts for edge cases where no employees fall within the specified range, ensuring a valid numeric output. The implementation filters employees whose ages fall within the given range, sums their salaries, divides by the number of employees, and returns 0 if no employees match the criteria.

3.7 Average Performance by Training Hours

The average department performance by training hours range metric tests the relationship between employee training investment and performance outcomes. This supports evaluation of training effectiveness across the organisation. The implementation filters employees within the specified training hours range, groups

them by department, calculates the average performance rating for each group, and assigns a value of 0 where no data exists.

4. Testing and Validation

Testing was conducted using Pytest to verify the correctness of all functions. Each function was tested independently and validated against expected outputs. Edge cases such as empty datasets and invalid values were handled. All tests passed successfully.

5. Conclusion

The project applied fundamental Python techniques to clean and analyse HR data. Data cleaning ensured the dataset was consistent and reliable, preventing errors during analysis. (Osborne, 2013) Aggregation was then used to compute key metrics such as retention, turnover, and performance. All functions followed the required constraints and were validated through testing, a process critical for ensuring the integrity of computational results (Wickham, 2014). The project demonstrates how structured data processing leads to accurate and meaningful insights.

Referencing

Osborne, J. W. (2013). Best practices in data cleaning: A complete guide to everything you need to do before and after collecting your data. SAGE Publications.

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